Experiment 05 — Spectral Lag and Latent Boundary Corrections

Two-Layer CTMC experimental framework

DEVELOPMENT RESULTS — NOT FINAL

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Abstract

This experiment tests two candidate aggregate-descriptor corrections to premature inter-community leakage in MicroMacro simulation. A spectral lag delays source contagiousness, while a latent-boundary state relaxes toward aggregate prevalence. Full Micro is the reference and all five variants share the network and initial condition within a replication. Inference uses the replication as the independent unit and distinguishes completed-only Wasserstein metrics from censoring-aware all-run RMST metrics. The present document reports the validation stage. These development estimates are diagnostic and are not final scientific conclusions.

1 Research question

Can spectral lag, latent-boundary relaxation, or their combination reduce sparse chain-of-cliques terminal-arrival bias without knowing bridge-node identities?

2 Motivation

Section 3.7 of *Notes on Micro–Macro Epidemic Networks* reports premature leakage in the sparse benchmark. Experiment 01 separated hazard and placement effects but left bridge-specific error. The corrections tested here are *candidate corrections proposed within the experimental framework, motivated by spectral epidemic-threshold bounds*; they are not attributed to the article.

3 Experimental design

The five paired variants are V0 Full Micro, V1 production aggregate MicroMacro, V2 lag-shifted MicroMacro, V3 latent-boundary MicroMacro, and V4 combined. The baseline has $L = 10$, $n = 100$, $k = 5$, mean internal degree approximately 6, $\beta_{in} = \beta_{out} = 0.003$, $\gamma = 0$, $\tau_{micro} = 1$, and $T_{end} = 4000$. Mini-sweeps vary $k \in \{1, 5, 10\}$ and $d \in \{4, 6, 10\}$ without taking a Cartesian product. Network and initialization streams are shared within replication; event streams are variant-specific.

4 Mathematical definitions

For community i, N_i is its size, $i_i(t)$ its infected count, $\bar{i}_i(t) = i_i(t)/N_i$, and $\bar{s}_j(t)$ is normalized susceptibility. The reference macro hazard is

$$\lambda_{ij}^{ref}(t) = \beta_{out} W_{ij} \bar{i}_i(t) \bar{s}_j(t).$$

The lag is

$$\tau_{find}(i) = \frac{\ln N_i}{\beta_{in}(\lambda_1(G_i) - 1)}, \qquad \bar{i}_i^{lag}(t) = \bar{i}_i(\max(0, t - \tau_{find}(i))),$$

and $\lambda_{ij}^A = \beta_{out} W_{ij} \bar{i}_i^{lag} \bar{s}_j$. The latent rate and analytical synchronization update are

$$\kappa(i) = \frac{\beta_{in}(\lambda_1(G_i) - 1)}{\ell(G_i)}, \qquad \bar{i}_i^B(t + \tau) = \bar{i}_i^B(t) + (1 - e^{-\kappa(i)\tau})(\bar{i}_i(t) - \bar{i}_i^B(t)),$$

with $\bar{i}_i^B(0) = 0$ and $\lambda_{ij}^B = \beta_{out} W_{ij} \bar{i}_i^B \bar{s}_j$. V4 applies the same latent update to $\bar{i}_i^{lag}$. If $\lambda_1 \leq 1$ or ℓ is undefined, corrected outgoing hazard is zero and the descriptors are flagged degenerate. Arrival definitions are $T_i = \inf\{t : i_i(t) > 0\}$, $T_{last} = T_{L-1}$, and $\Delta T_i = T_{i+1} - T_i$; censored values equal T_{end} with a separate observation flag.

5 Parameters

Each configured scenario uses a connected fixed-edge-count Erdős–Rényi community, uniform susceptible import placement, and one uniformly selected seed in community zero. Graph descriptors are computed once before simulation.

6 Implementation details

Adjacency spectral radius is computed by shifted power iteration to tolerance 10^{-6} with at most 1000 iterations and a residual-norm convergence check. Mean shortest-path length is computed on the unweighted local graph. Lag history uses fixed synchronization points and linear interpolation. The latent ODE uses the analytical exponential update, not Euler integration. No descriptor fallback is permitted. Outputs retain deterministic stream metadata and per-replication graph checksums. Additional diagnostics have distinct purposes: integrated hazard suppression measures how strongly each correction changes V1 exposure; synchronized correlation between $\bar{i}_i$ and the latent state checks relaxation semantics; and V0 time-to-first-boundary versus τ_{find} assesses the spectral lag as a descriptor without feeding bridge identity into a corrected variant.

7 Sanity checks

Table 1: Recorded stage gates and actual diagnostic values.

Gate	Value	Status
pairing	groups=100; failures=0	passed
snapshot_parquet_metadata	{"failures": 0, "files_checked": 500}	passed
snapshot_raw_sha256	{"failures": 0, "files_checked": 500}	passed
source_writer_isolated	snapshot contains checkpoint summaries only; active raw/checkpoints are not modified	passed

Interpretation below is conditional on the formula, reproducibility, pairing, reference, schema, and production-immutability gates recorded in the manifest.

8 Results tables

Table 1 labels W_1 as completed-only and RMST/completion as all-run metrics.

Scenario	Variant	rel. W_1	ΔRMST [95% CI]	Completion	Status
baseline	Full Micro	0	0 [0, 0]	0.99	ok
baseline	Baseline MicroMacro	0.06939	200.6 [98.25, 307.7]	0.98	ok
baseline	Lag	NA	1071 [995.9, 1152]	0	no_completed_arrivals
baseline	Latent boundary	0.2421	905.8 [807.4, 989.9]	0.44	ok
baseline	Lag + latent	NA	1071 [991, 1146]	0	no_completed_arrivals

9 Figures

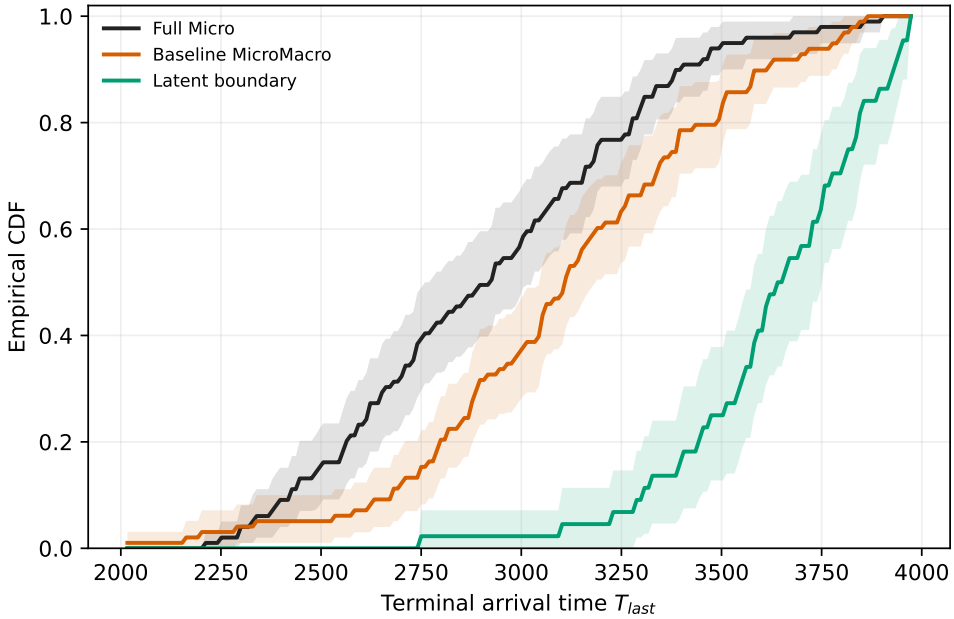


Figure 1: Baseline empirical CDF of completed terminal arrivals. Bands are pointwise 95% replication-bootstrap intervals; the statistical unit is a replication.

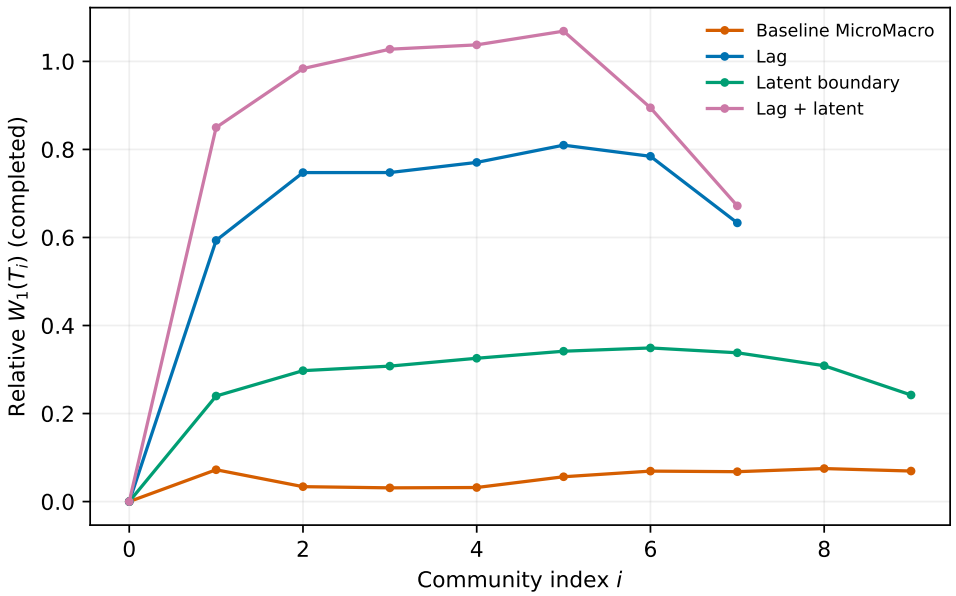


Figure 2: Layer-wise completed-only relative $W_1(T_i)$ against V0. The statistical unit is a replication.

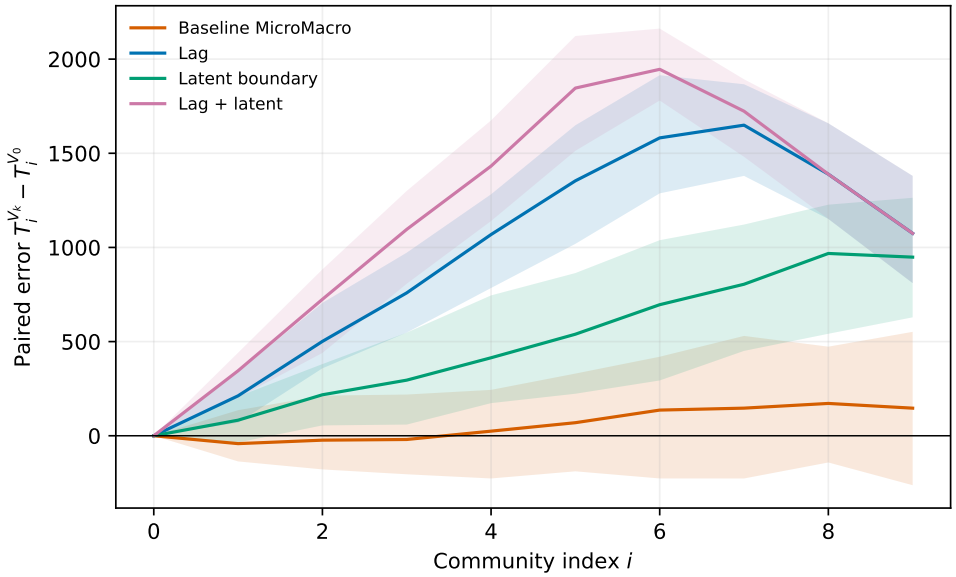


Figure 3: Paired cumulative arrival error (variant minus V0), median and IQR over replications; censored arrivals retain the common horizon.

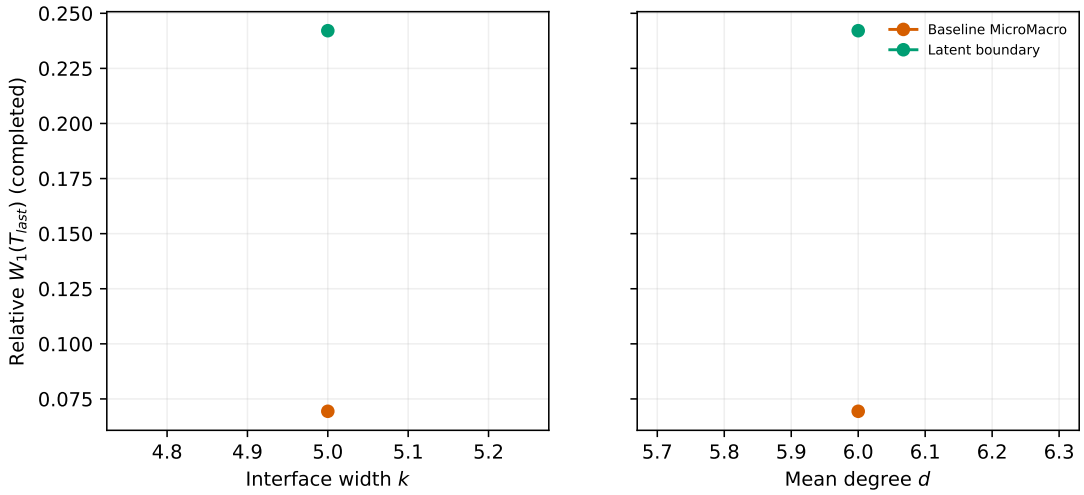


Figure 4: Completed-only terminal relative W_1 across the interface-width and density mini-sweeps. Baseline supplies the shared $k = 5, d = 6$ point.

10 Uncertainty analysis

All confidence intervals use 1000 paired bootstrap resamples of replication IDs. Event rows are never bootstrap units. W_1 compares completed terminal-arrival distributions and is normalized by the V0 completed mean. ΔRMST is variant minus V0 using all horizon-restricted runs. Equivalence requires the entire ΔRMST interval to lie within $\pm 0.05\,\mathrm{mean}(T_{last}^{V0})$.

11 Scientific interpretation

H1 is assessed by V2 versus V1, H2 by V3 versus V1, and their joint behavior by V4. The replication-level factorial interaction $V4 - V2 - V3 + V1$ on restricted terminal arrival is -705.2 (95% CI -799.8 to -613.3). Claims of improvement require both passed gates and the reported uncertainty intervals; an interval containing zero alone is not evidence of equivalence. In the baseline validation sample, V1 completed in 98% of runs with relative completed-only $W_1 = 0.06939$. H1 (V2) and H1+H2 (V4) each had zero terminal completions, so their terminal W_1 values are undefined; their all-run ΔRMST values were 1071 and 1071. H2 (V3) completed in 44% of runs, with relative completed-only $W_1 = 0.2421$ and all-run ΔRMST = 905.8. Thus none of the literal corrections improved the validation outcome over V1; this development sample does not confirm H1, H2, or monotone additivity.

12 What the experiment can identify

It can estimate relative bias reduction in the sparse baseline, distinguish H1 from H2 within the factorial cells, assess their interaction, and describe limited scaling across the specified d and k values.

13 What the experiment cannot identify

It cannot identify recovery effects, synchronization-interval effects, placement effects, generalization beyond connected ER communities, or computational speedup. Runtime is an implementation diagnostic only.

14 Limitations

Lag may interact with midpoint hazard integration, and latent relaxation may interact with the exponential-clock residual update. The spectral formula is a candidate scaling law over a small descriptor range. Completed-only Wasserstein statistics condition on arrival and therefore must be read beside completion and all-run RMST.

15 Unresolved questions

The current design does not determine whether any remaining discrepancy is caused by synchronization timing, placement, or non-ER boundary geometry.

16 Recommended next experiment

No recommendation is made outside the staged Experiment 05 evidence. Any follow-up specified here after validation must preserve pairing and separately vary only one previously fixed mechanism.

17 Exact reproduction commands

```
python -m two_layer_ctmc.experiments.experiment_05_spectral_corrections.cli \
  --stage smoke --scenario baseline --replications 5 --master-seed 270000
python -m two_layer_ctmc.experiments.experiment_05_spectral_corrections.cli \
  --stage validation --scenario all --replications 20 --resume --master-seed 270000
# Run only after explicit approval:
python -m two_layer_ctmc.experiments.experiment_05_spectral_corrections.cli \
  --stage final --scenario all --replications 100 --resume \
  --validation-approved --master-seed 270000
```